

Table 1: Criteria for Classification of Attribute Coverage in the Review-of-Reviews Analysis

Attribute	× Not Covered	△ Partially Covered	✓ Clearly Covered
1. AMR Coverage Extent to which the review addresses antimicrobial resistance (AMR) as a central research domain.	AMR is not meaningfully addressed.	AMR is included but represents only one part of a broader review, or coverage is relatively limited.	AMR is a major focus, including AMR prediction, detection, resistance mechanisms, or antimicrobial susceptibility.
2. Multiple Pathogens / Microbial Classes Whether the review considers AMR across more than one pathogen, bacterial species, or microbial class.	The review is restricted to a single pathogen/species.	More than one pathogen appears, but coverage is limited or not systematically compared.	Multiple pathogens/species/classes are explicitly analyzed or compared.
3. Multiple Data Modalities Whether the review considers more than one type of input/data source (e.g., WGS, transcriptomics, proteomics, mass spectrometry, EHR, imaging, phenotypic data).	Only one modality is considered.	More than one modality is mentioned but coverage is limited.	Multiple distinct modalities are explicitly reviewed.
4. Multimodal Integration Whether the review addresses integration of two or more data modalities within a model/framework, rather than discussing modalities separately. Note: multiple modalities $\neq$ multimodal integration.	Different modalities may be discussed separately, but integration is not addressed.	Multimodal approaches are mentioned or examples provided, but integration is not systematically examined.	Multimodal integration is explicitly analyzed, compared, or represented through integrated modelling approaches.
5. Prediction / ML Whether AI, machine learning, deep learning, or related computational approaches are used/reviewed for AMR prediction or classification.	AI/ML prediction is not meaningfully addressed.	AI/ML is discussed but represents only a limited component.	AI/ML prediction is a major component of the review.
6. XAI / Explainability Whether the review addresses interpretability, explainability, feature attribution, or techniques intended to explain AI/ML predictions.	No meaningful discussion of explainability.	Explainability is mentioned or discussed only in limited contexts.	XAI/explainability is explicitly analyzed as a substantial research aspect.

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7. XAI Methods Whether the review identifies and discusses specific methods used to explain AI/ML models (e.g., SHAP, LIME, Integrated Gradients, saliency maps, attention, feature importance).	No specific XAI methods are discussed.	XAI is discussed but methods are only briefly or incompletely described.	Specific XAI methods are identified and discussed.
8. XAI Method Comparison Whether the review compares different XAI approaches with respect to application, strengths, limitations, validation, or suitability. Note: simply listing methods does not qualify as covered.	XAI methods are not compared.	Multiple methods are listed or categorized, but systematic comparison is limited.	Multiple XAI methods are explicitly compared.
9. Explanation Quality Whether the review evaluates whether generated explanations are meaningful, useful, valid, or sufficiently informative.	No meaningful discussion of explanation quality.	Explanation quality is mentioned or indirectly assessed, but no comprehensive evaluation is provided.	Explanation quality is explicitly evaluated or discussed using identifiable criteria/evidence.
10. Explanation Faithfulness Whether the explanation actually reflects the model’s reasoning or the contribution of highlighted features to the prediction. Note: biological plausibility $\neq$ proven model faithfulness.	Faithfulness is not discussed.	Faithfulness-related concepts are addressed indirectly (e.g., attribution validation, biological consistency), but no dedicated faithfulness framework is used.	Faithfulness is explicitly evaluated using identifiable methodology or evidence (e.g., perturbation tests, feature removal).
11. Explanation Stability Whether explanations remain consistent when the model, dataset, samples, random seed, or other conditions change.	Stability is not discussed.	Stability is mentioned or partially assessed but without systematic evaluation.	Stability/consistency is explicitly evaluated (e.g., across runs or similar samples).
12. Explanation Reproducibility Whether the same model/data/conditions produce reproducible explanations (distinct from general research reproducibility).	Explanation reproducibility is not addressed.	Reproducibility of XAI results is mentioned but not systematically evaluated.	Explanation reproducibility is explicitly assessed.
13. AMR-specific XAI Evaluation Framework Whether the review proposes, applies, or systematically discusses an evaluation framework specifically suited to assessing XAI in AMR applications.	No AMR-specific XAI evaluation framework is provided or discussed.	AMR-specific evaluation criteria/categories are discussed, but no comprehensive standardized framework is established.	A comprehensive AMR-specific framework or structured evaluation protocol is explicitly provided.

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14. Biological / Mechanistic Grounding Whether model predictions or explanations are connected to known biological mechanisms of AMR (e.g., resistance genes, mutations, pathways, proteins).	No meaningful biological/mechanistic interpretation.	Biological interpretation is mentioned or partially connected to known mechanisms.	Biological/mechanistic interpretation is explicitly examined or validated.
15. Genotype–Phenotype Relationship Whether the review examines the relationship between genetic/molecular features and observed AMR phenotypes.	No meaningful discussion.	The relationship is mentioned but not deeply investigated.	The genotype–phenotype relationship is explicitly discussed or analyzed.
16. Clinical Validation Whether AI/ML findings are evaluated using clinical data, clinical cohorts, or clinically relevant outcomes. Note: using EHR data $\neq$ clinical validation.	No clinical validation.	Clinical relevance/clinical cohorts are discussed, but rigorous clinical validation is limited.	Explicit clinical validation is performed or systematically reviewed.
17. External Validation Whether models are evaluated on data independent from the data used for model development (e.g., independent hospital, cohort, dataset, or site).	No external validation is reported/discussed.	External validation is mentioned as a need or occurs only in limited cases.	External validation is explicitly performed or systematically examined.
18. Generalizability Whether the review examines how well findings/models transfer across pathogens, populations, hospitals, geographic regions, or clinical settings.	Generalizability is not meaningfully addressed.	Generalizability is discussed but only partially tested/evaluated.	Generalizability is explicitly analyzed.
19. Reproducibility Whether the review addresses the ability of researchers to reproduce the computational research or model results (open data, code, pipelines). Distinct from explanation reproducibility.	Reproducibility is not addressed.	Some reproducibility-related aspects are discussed, but coverage is incomplete.	Reproducibility is explicitly addressed using one or more substantive criteria.

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20. Standardization Whether the review addresses the need for or availability of standardized procedures, datasets, evaluation metrics, or reporting practices.	Standardization is not meaningfully discussed.	Standardization issues are mentioned but only for limited aspects.	Standardization is explicitly and substantially discussed across one or more important methodological dimensions.
21. AMR-specific Multi-modal XAI Whether the review specifically addresses the combination of AMR + multimodal data + AI/ML + explainability together. Note: multiple modalities + XAI does not automatically equal this category.	Neither multimodal AMR modelling nor its explainability is meaningfully addressed together.	The review discusses multimodal AMR approaches and XAI separately, or provides limited overlap examples, but does not systematically analyze multimodal XAI.	The review explicitly examines multimodal AMR models and their explainability together.